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### Rotor For Axial Flux Motor And Method Of Manufacture

#### Abstract

A rotor for an axial flux motor and a method of manufacture includes a rotor core having a rotor hub with a plurality of spaced apart fingers radially extending from the rotor hub and forming a magnet pocket between adjacent fingers and into which a magnet is inserted, where the rotor core is formed of a plurality of layers of material arranged into a layer stack, with each layer having a set of primary fibers extending in a select direction that is a different direction from the set of primary fiber layers of an adjacent layer in the layer stack.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of U.S. Non-Provisional application Ser. No. 17/667,421, filed on Feb. 8, 2022, which claims priority to U.S. Provisional Application No. 63/147,090, filed on Feb. 8, 2021, the benefit of which is claimed and the disclosures of which are incorporated herein by reference in their entirety.

### FIELD OF THE DISCLOSURE

[0002] The present invention relates in general to electric motors, and particularly to axial flux motors wherein a rotor and a stator are coaxially arranged. More specifically, the present invention relates to a rotor used in such axial flux motor and a method of producing the rotor.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Various embodiments of the present disclosure will be understood more fully from the detailed description given below and from the accompanying drawings of various embodiments of the disclosure. In the drawings, like reference numbers may indicate identical or functionally similar elements.

[0004] FIG. 1 is an exploded perspective view of an axial flux motor of the disclosure.

[0005] FIG. 2A is a perspective view of one embodiment of a rotor assembly.

[0006] FIG. 2B is an exploded perspective view of the rotor assembly of FIG. 2A.

[0007] FIG. 2C is an enlarged perspective view of a portion of the rotor core illustrating the layered construction of the rotor core.

[0008] FIG. 3A is a plan view of one layer of rotor core illustrating a first fiber direction of the layer.

[0009] FIG. 3B is an exploded view of one embodiment of the rotor core in which adjacent layers have different fiber directions.

[0010] FIG. 4A illustrates a method and device for manufacturing the rotor assembly of FIGS. 2 and 3.

[0011] FIG. 4B illustrates a mold utilized in formation of embodiments of the rotor assembly described herein.

[0012] FIG. 4C illustrates insertion of the retaining ring in a mold during manufacturing the rotor assembly of FIGS. 2 and 3.

[0013] FIG. 4D illustrates insertion of the magnets in a mold during manufacturing the rotor assembly of FIGS. 2 and 3.

[0014] FIG. 4E illustrates insertion of the rotor core (spider stack) in a mold during manufacturing the rotor assembly of FIGS. 2 and 3.

[0015] FIG. 5 is a plan view of a rotor assembly of the disclosure.

[0016] FIG. 6A is an enlarged elevation view of a magnet in a magnet pocket of the rotor assembly of FIG. 5.

[0017] FIG. 6B is an enlarged elevation view of the magnet of FIG. 6A with a spacer positioned radially outward from the magnet.

[0018] FIG. 6C is an enlarged elevation view of a grooved magnet in a grooved magnet pocket with the magnet spaced apart from the spider edges.

[0019] FIG. 6D is an enlarged elevation of the grooved magnet of FIG. 6C illustrating a magnet retaining member inserted between the magnet and the spider.

[0020] FIG. 7 is an enlarged elevation of wedge-shaped spider fingers of the rotor core enabling rectangular shaped magnets.

[0021] FIG. 8 is a plan view of a rotor core assembly having spacers inserted between the magnets and the outer ring.

[0022] FIG. 9A is a perspective view of another embodiment of a rotor assembly.

[0023] FIG. 9B is an exploded view of the rotor assembly of FIG. 9A.

[0024] FIG. 9C is a cut-away elevation view of the rotor assembly of FIG. 9A.

[0025] FIG. 9D is a perspective view of the partially assembled rotor assembly of FIG. 9A.

[0026] FIG. 9E is an enlarged perspective view of the connection between magnets and fingers of the rotor assembly of FIG. 9A.

[0027] FIG. 10A is a perspective view of an embodiment of a rotor assembly with stress relief spacers between each magnet and the outer ring.

[0028] FIG. 10B is an exploded view of the rotor assembly of FIG. 10A.

[0029] FIG. 11A is a perspective view of another embodiment of a rotor assembly with stress relief spacers between each magnet and the outer ring.

[0030] FIG. 11B is an exploded view of the rotor assembly of FIG. 11A.

[0031] FIG. 11C is an enlarged perspective view of the connection between magnets and fingers of the rotor assembly of FIG. 11A.

[0032] FIGS. 11D and 11E are perspective views of a connector assembly utilized in the rotor assembly of FIG. 11A.

[0033] FIG. 12A is a partial assemble perspective view of another embodiment of the rotor assembly of FIG. 11C but with alternative connector assemblies.

[0034] FIGS. 12B and 12C are perspective views of a connector assembly utilized in the rotor assembly of FIG. 12A.

[0035] FIG. 12D is a partial assemble perspective view of another embodiment of the rotor assembly of FIG. 11C but with alternative connector assemblies.

[0036] FIGS. 12E and 12F are perspective views of a connector assembly utilized in the rotor assembly of FIG. 12D.

[0037] FIG. 13A is a perspective view of an embodiment of a rotor assembly with trapezoidal shaped magnets and magnet pockets.

[0038] FIG. 13B is plan view of the rotor assembly of FIG. 13A.

[0039] FIG. 13C is an exploded view of the rotor assembly of FIG. 13A.

[0040] FIG. 14A is an exploded view of another embodiment of a rotor assembly of the disclosure.

[0041] FIG. 14B is a perspective view of the rotor assembly of FIG. 14A.

[0042] FIG. 14C is a partial cut-away assembly view of the rotor assembly of FIG. 14A illustrating cooling channels.

[0043] FIG. 14D is a cut-away plan view of a rotor finger of the rotor assembly of FIG. 14A.

#### DETAILED DESCRIPTION OF THE EMBODIMENTS

[0044] Disclosed herein is a rotor assembly for an axial flux motor where the rotor core is formed of plurality of fiber layers stacked and bonded together. Each layer of the stack is oriented so that the primary fibers of the layer extend in a select direction and wherein the sets of primary fibers of adjacent fiber layers extend in different directions. Each layer is cut so as to have the shape of a rotor hub with a plurality of radially extending, spaced apart fingers so that the stacked and bonded layers form magnet pockets between adjacent fingers. A magnet is positioned in each magnet pocket of the rotor core. In one or more embodiments, the fingers of the rotor core may be triangular or wedge shaped, thereby creating rectangular magnet pockets for receipt of rectangular magnets. A radial spacer may be inserted in the magnet pocket radially outward from the magnet to reduce stress on the magnet, especially stress that might be created by a radial retaining ring. Likewise, to reduce magnet stresses, each magnet may be grooved along at least its side edges where hardened resin or offset connectors in the form of pins or blades may be utilized to center the magnet in the pocket so as to be spaced apart from the fingers. In some embodiments, radial spacers and offset connectors may be integrally formed. In one or more embodiments, the magnet

pockets may be trapezoidal in shape tapering outward so that the fingers radially retain like-shaped magnets in the magnet pockets. In one or more embodiments, air flow channels may be formed along fingers and between magnets to enhance cooling. In one or more embodiments, a retainer cover may be disposed on either side of the rotor core to axially secure magnets in the pockets. In this regard, one or both covers may further include an outer radial lip that also secures magnets radially. The retaining cover may include integrally formed magnet seats and/or air flow channels. [0045] Referring to FIG. 1, there is shown an axial flux type electric motor assembly **100**. Electric motor assembly **100** includes at least one rotor assembly **110** and at least one stator assembly **112** disposed axially from one another along a motor axis **114**. In the illustrated embodiment, a rotor spindle or shaft **116** extends along axis **114** and supports rotor assembly **110**. Rotor shaft **116** may include a spindle flange **117** that attaches to rotor assembly **110**. Rotor shaft **116** may in turn be supported by one or more bearings **118**.

[0046] Although only a single stator assembly **112** may be utilized, in the illustrated embodiment, two stator assemblies **112a**, **112b** are shown and positioned on opposing sides of rotor assembly **110** along axis **114**.

[0047] A motor housing **120** and opposing stator support or end plates **122** enclose rotor assembly **110** and the one or more stator assemblies **112**. In one or more embodiments, at least one end plate **122** supports a stator assembly **112** on an interior surface **123** of the end plate **122**. In the illustrated embodiment, end plate **122a** supports stator assembly **112a** and end plate **122b** supports stator assembly **112b**.

[0048] To the extent an end plate **122** supports a stator assembly **112**, the end plate **122** may include cooling a cooling mechanism **124** positioned along the exterior surface **126** of the end plate **122**.

[0049] In one or more embodiments, cooling mechanism **124** may form one or more coolant flow paths **128** along the exterior surface **126** of end plate **122**. Coolant flow path **128** may be one or more cooling channels **130** formed in the exterior surface **126** of end plate **122**.

[0050] In the illustrated embodiment, a plurality of fluidically connected cooling channels **130** are illustrated and are generally positioned to extend around the end plate **122** opposite the stator assembly **112** mounted on the interior surface **123** of end plate **122**. Although not limited to a particular configuration, in one embodiment, cooling channels **130** may form star shape along the exterior surface **126** to maximize cooling while allowing fasteners **127** to secure various motor components to end plate **122** from the interior surface **123** without extending through end plate **122** to the exterior surface **126** of end plate **122**. In other words, threaded bores for engagement by fasteners **127** may be formed on the interior surface **123** of end plate **122**, but the bores do not extend all the way through end plate **122**. It will be appreciated that such an arrangement minimizes the likelihood that fluid within the interior of motor assembly **100** could leak or migrate out or that vice-versa. In other embodiments, coolant flow path **128** may be formed of tubing (not shown) positioned on the exterior surface **126**. In yet other embodiments, coolant flow path **128** may be formed of ribs or fins (now shown) extending from exterior surface **126**, while in other embodiments, coolant flow path **128** may simply be a coolant chamber formed between end plate **122** and an outer plate **134**. Notwithstanding the foregoing, it will be appreciated that cooling channels **130** formed in the exterior surface **126** of end plate **122** may be particularly desirable to minimize the overall axial length of electric motor assembly **100**. In any event, one or more ports **125** in fluid communication with flow path **128** may be utilized to introduce and extracted coolant from the from cooling mechanism **124**.

[0051] Each stator assembly **112** is generally formed of a stator core **138** and stator windings **140** as may be known to persons of skill in the art. In this regard, stator windings **140** may be formed of electric wire. The disclosure is not limited to a particular configuration for stator assembly **112**.

[0052] With reference to FIGS. 2A and 2B and ongoing reference to FIG. 1, rotor assembly **110** is generally formed of a rotor core or yoke **142** disposed to carry a plurality of magnets **144**. In one or more embodiments, rotor core **142** is spider shaped and formed of a hub **146** from which fingers

**148** radially extend. Fingers **148** are spaced apart from one another around the perimeter of hub **146** so as to form magnet pockets **150** between adjacent fingers **148**. In one or more embodiments, rotor assembly **110** further includes a rotor ring **152** disposed radially outward from fingers **148**. In one or more embodiments, rotor ring **152** and rotor core **142** may be separate components, such as is shown in FIGS. 1, 2A and 2B, while in other embodiments, rotor ring **152** and rotor core **142** may be integrally formed.

[0053] In one or more embodiments, rotor core **142** is formed of a non-conducting composite material. Similarly, in one or more embodiments, rotor ring **152** is formed of a non-conducting composite material.

[0054] Turning to FIGS. 2A-2B, embodiments of a rotor assembly **110** are shown and generally described as rotor assembly **210**. Rotor assembly **210** generally includes a rotor core or yoke **242** disposed to carry a plurality of magnets **244**. In one or more embodiments, rotor core **242** is spider shaped and formed of a hub **246** from which fingers **248** radially extend. Fingers **248** are spaced apart from one another around the perimeter of hub **246** so as to form magnet pockets **250** between adjacent fingers **248**. Rotor assembly further includes a rotor ring **252** disposed radially outward from fingers **248**. Although not limited to a particular number of fingers **248** and pockets **250**, in the illustrated embodiment, rotor core **242** includes ten fingers **248** and ten magnet pockets **250**.

[0055] In the illustrated embodiment of rotor assembly **210**, fingers **248** are generally rectangular in shape so that magnet pockets **250** are generally wedge shaped.

[0056] As such, in this illustrated embodiment, magnets **244** are generally wedge shaped, where each magnet **244** has a radially outer edge **260** of a greater length than a radially inner edge **262** which edges **260**, **262** are joined by side edges **264**. In one or more embodiments, as best seen in FIG. 2B, at least one and preferably each side edge **264** of a magnet **244** includes a groove or slot **266** extending along at least a portion of the length of side edge **264** between radially outer edge **260** and radially inner edge **262**. In one or more embodiments, each groove **266** fully extends between radially outer edge **260** and radially inner edge **262**, while in other embodiments groove **266** extends from radially outer edge **260** and is spaced apart from radially inner edge **262**. In yet other embodiments, groove **266** is spaced apart from radially outer edge **260** and extends from radially inner edge **262**. Finally in other embodiments, groove **266** is formed along side edge **264** to be spaced apart from both radially outer edge **260** and radially inner edge **262**.

[0057] Similarly, each finger **248** of hub **246** has a side edge **268** that extends from a proximal end **269a** adjacent the hub **246** to a distal end **269b** at the periphery of the rotor core **242**. In one or more embodiments, the side edges **268** of each finger may be generally smooth without any feature formed therealong. In one or more other embodiments, each side edge **268** of a finger **248** includes a groove or slot **270** extending along at least a portion of the length of side edge **268** between the proximal end **269a** and the distal end **269b** of finger **248**.

[0058] In one or more embodiments, each groove **270** fully extends between the proximal end **269a** and the distal end **269b** of finger **248**, while in other embodiments groove **270** extends from distal end **269b** and is spaced apart from proximal end **269a**. In yet other embodiments, groove **270** is spaced apart from distal end **269b** and extends from proximal end **269a**. Finally in other embodiments, groove **270** is formed along side edge **268** to be spaced apart from both the distal end **269b** and the proximal end **269a** of finger **248**.

[0059] In one or more embodiments, rotor ring **252** is a pre-formed ring that is positioned around rotor core **242**. In some embodiments, rotor ring **252** is a continuous, solid ring. In this regard, rotor ring **252** may be formed of a composite material, such as a fiber material. Rotor ring **252** has a radially inner edge **276** and a radially outer edge **278**, where radially inner edge **276** may abut fingers **248** when positioned around rotor core **242**. In one or more embodiments, the radially inner edge **276** of solid, pre-formed rotor ring **252** is formed. As will be appreciated, each of the carbon fiber material and glass fiber material may be formed of a substrate on which the fibers are carried.

[0060] Turning to FIG. 2C, in one or more embodiments, rotor core or yoke **242** may be formed of

a multiplicity of layers **280** of fiber material arranged in a stack **282** or “book” as shown. Each fiber layer **280** may be a material having a set of primary fibers.

[0061] Turning to FIG. 3A and with ongoing reference to FIG. 2C, a layer **280** of the fiber material that forms stack **282** is illustrated in more detail. Specifically, as shown, each layer **280** may be characterized as having at least a primary set of fibers **284** extending in a first direction, as generally indicated by the arrow **286**. Fibers **284** need not be limited to a particular type of fiber, but may include glass fiber, carbon fiber or other fibers known in the art. In one or more embodiments, the fibers may be short fibers or long fibers so long as the primary fibers are generally aligned in a distinct direction. Likewise, the material carrying the fibers need not be limited to a particular type of material. In one or more embodiments, the material may be cloth or woven. In other embodiments, the material may be polymer material, such as a sheet of plastic. Each layer **280** may further include one or more secondary sets of fibers **287**, such as may be arranged in a cross-weave. However, it is the primary set of fibers that are utilized in the assembly of the stack **282** of rotor core **242**, as is best illustrated in FIG. 3B. In one or more embodiments, each layer **280** may be formed by punching the layer **280** from a sheet of fiber material. In other embodiments, each layer **280** may be cut from a sheet of fiber material, such as through precision laser cutting or the like. Although not a requirement, in one or more embodiments the fiber material sheet may be pre-impregnated with a bonding material or adhesive for use in manufacturing hub **246**. Likewise, in one or more embodiments, to assist in manufacturing as described herein, an indicator, such as arrow **286**, may be stamped or otherwise imprinted on the layer **280** at the time it is cut or punched in order to indicate the direction of the primary set of fibers **284**.

[0062] In FIG. 3B, a partially exploded view of stack **282** is shown. In each case, as individual layers **280** are formed into the stack **282**, the layer **280** is rotated through a predetermined angle so that adjacent layers **280** have their primary sets of fibers **284** extending in different directions. It has been found that by arranging layers as described in the ordered directions as described herein, the performance life of rotor assembly **210** can be significantly enhanced. Thus, as shown, a first layer **280a** has the primary sets of fibers **284** extending in a first direction as indicated by arrow **286a**, and the adjacent second layer **280b** has the primary sets of fibers **284** extending in a second direction as indicated by arrow **286b**. Likewise, a third layer **280c** has the primary sets of fibers **284** extending in a third direction as indicated by arrow **286c**, and the adjacent fourth layer **280d** has the primary sets of fibers **284** extending in a fourth direction as indicated by arrow **286d**. In preferred embodiments, this rotational sequence of layers **280** continues as the stack **282** is formed. Thus, a fifth layer **280e** adjacent the fourth layer **280d** is therefore arranged so that the primary sets of fibers **284** of the fifth layer **280e** extends in a fifth direction as indicated by arrow **286e**.

[0063] It will be appreciated that while it is desirable to maximize the number of directions **286** of layers **280** in stack **282**, in one or more embodiments, the plurality of layers **280** are at least arranged so that sets of fibers **284** extend in at least three different directions with successive or adjacent layers extending in different directions.

[0064] In one or more embodiments of a stack **282** having three directions **286**, successive layers are rotated sixty degrees from one another. It will be appreciated that the number of fingers **248** may determine degree of rotation of each layer **280**. Thus, in one or more embodiments, the number of different directions **286** for the sets of fibers **284** is equivalent one half the number of fingers **248** to the extent fibers are generally uniform in cross-section or where fibers are not uniform in cross-section (such that a fiber extending in one direction does not have the same material properties as extending 180 degrees in the opposite direction). Thus, in other embodiments, the number of different directions **286** for the sets of fibers **284** is equivalent the number of fingers **248**. In the embodiment illustrated in FIG. 3B, it will be appreciated that five directions **286** are illustrated based on the ten fingers **248** of the rotor core **242**. Thus, a sixth layer (not shown) that is adjacent the fifth layer **280e**, would have fibers **284** of the sixth layer that extend in the first direction as indicated by arrow **286a**. In one or more embodiments of a stack **282**

having five directions **286**, successive layers are rotated 18 degrees from one another. In one or more embodiments, successive layers are rotated sixty degrees from one another. While uniform angular rotations of successive layers are used in the example, it will be appreciated that the disclosure is not limited to any particular angular rotation of successive layers, if any rotation at all. [0065] During the assembly of rotor core **242**, it will be appreciated that each layer **280** may be machine cut, stamped or otherwise formed from a fiber cloth or bolt of fiber material. In the case of each layer **280**, in one or more embodiments, the direction of the primary set of fibers **284** for is marked or otherwise indicated by a reference point so that each layer **280** may be oriented as described above relative to successive adjacent layers. It will also be appreciated that each individual layer **280** may be formed so that when combined into stack **282**, various three dimensional shapes or features may be present, such as slots **270**.

[0066] While rotor ring **252** (see FIGS. 2A and 2B) is preferably a solid, pre-formed ring that is positioned around rotor core **242**, in one or more other embodiments, rotor ring **252** may also be formed of layers of fiber material formed into a ring stack as generally described above with respect to rotor core **242**. In such case, in preferred embodiments, a multiplicity of continuous rings may be cut from fiber material and formed into a stack. The ring layers are preferably continuous to ensure an even application of hoop stress about the ring, which is particularly desirable given the high rotational speeds to which the rotor assembly **210** will be subjected during use. Consecutive ring layers may be angularly oriented relative to one another so that fiber direction is offset from adjacent layers of fiber material as described. In one or more embodiments, a layer **280** may be cut to include a ring such that the ring and fingers of the layer **280** are integrally formed.

[0067] In FIG. 4A, a method **400** of manufacturing rotor assembly **210** is illustrated. As described above, a first step **402** in method **400** is to cut a multiplicity of layers **280** from a fiber material. The fiber material is selected to have at least a primary fiber extending in an identifiable primary fiber direction. In one or more embodiments of step **402**, once a layer **280** is cut, the layer may be marked with a label, printed indicator or other device that provides an indication of the direction of the primary fiber. In one or more embodiments, the material is pre-impregnated with adhesive or a bonding material. In step **404**, the multiplicity of layers **280** are arranged in a stack **282** so that fingers **148** align. In one or more embodiments of step **404**, each layer is axially rotated relative to the adjacent layer so that the primary fibers **284** of one layer **280** are angularly offset from the primary fibers **284** of an adjacent layer **280**.

[0068] In one or more embodiments, the material forming each layer **280** is glass fiber material with glass fiber forming the primary fiber. In other embodiments, the material forming each layer **280** is carbon fiber material with carbon fiber forming the primary fiber. In other embodiments, other types of fiber may be utilized in the material forming each layer **280**.

[0069] In step **406**, once the stack **282** is formed, heat and/or pressure may be applied to the stack **282** in order to activate the adhesive or bonding material to form the composite rotor core **242**. In this regard, minimal heat and/or pressure may be applied only to the degree necessary to initially secure the individual layers **280** to one another, but without expressing the pre-impregnated adhesive or bonding material from the stack **282**. In one or more embodiments, a mold **300** such as that shown in FIGS. 4C-4E may be utilized to apply such initial heat and/or pressure to stack **282** in order to form rotor core **242**. Notably, in the prior art, rotor assemblies, and in particular the rotor core, are generally formed of milled metal or milled composite materials, as opposed to forming the rotor core **242** as described herein. In any event, mold **300** includes a mold cavity **301** with reliefs or other orienting features **302** to allow the stack **282** to be positioned within mold **300**.

[0070] In addition to utilizing mold **300** to form rotor core **242**, mold **300** may also be utilized to assemble the other components of rotor assembly **210** together, namely the rotor ring **252** and the magnets. **244**. In this regard, mold **300** may also include reliefs, projections or other orienting features **302** mentioned above, such as the illustrated magnet seats **302a** for receiving and positioning the magnets and a rotor core seat **302b** for receipt and alignment of stack **282** of rotor

core **242**. Notably, magnet seats **302a** may be a raised relief or depression relative to rotor core seat **302b**, depending on the features desired for rotor assembly **210** (such as cooling channels discussed below). In any event, in addition to initially bonding the layers **280** of stack **282** together to form the rotor core **242**, as will be described, the mold **300** may be used to activate the adhesive or bonding material to bond the rotor ring **252** and magnets **244** to the rotor core **242**. Notably, in one or more embodiments, the orienting features **302** are disposed so that magnets **244** are spaced apart from the fingers **248** leaving a gap so that the magnets **244** are not in physical contact with the fingers **248**. As will be appreciated, as bonding material is expressed during the formation steps below, the bonding material will flow into this gap and maintain the spaced apart relationship once manufacturing is complete, thereby minimizing stresses on the magnets during use of the rotor assembly **210**.

[0071] Thus, turning to FIG. **4C**, in step **408**, a solid, pre-formed rotor ring **252** may be positioned within the mold **300**. In one or more embodiments of step **408**, the rotor ring **252** is preheated prior to inserting it in mold **300**. Likewise, mold **300** may be preheated for receipt of rotor ring **252**. It will be appreciated that mold **300** includes mold cavity **301** with an outer diameter **306** disposed to receive and seat rotor ring **252**.

[0072] In step **410**, as illustrated in FIG. **4D**, magnets **244** are inserted into mold **300**. Each magnet **244** may be aligned on or in a magnet seat **302a**. In one or more embodiments, the magnets **244** are preheated prior to inserting it in mold **300**.

[0073] Finally, in step **412**, as illustrated in FIG. **4E**, stack **282** of layers **280** is inserted into mold **300**. As mentioned above, in one or more embodiments, stack **282** may be at least partially preassembled by applying enough heat or pressure to secure layers **280** to one another prior to insertion into mold **300** for step **412**. In other embodiments of step **412**, stack **282** may be simply inserted into mold **300** without this preassembly step. In this regard, it will be appreciated that unlike the rotor ring **252** and the magnets **244**, stack **282** is preferably not preheated so as to avoid expressing the pre-impregnated adhesive/bonding material prior to insertion into mold **300**. While the foregoing is the preferred order for inserting rotor assembly **210** components into mold **300**, the method of manufacture is not limited to a particular order. Notably, where the rotor ring **252** is formed of a stack of fiber material as described above, the rotor ring **252** is positioned in the mold **300** at the same time as the stack **282**, and in such case, the rotor ring **252** is not preheated.

[0074] In any event, in step **414**, once the rotor assembly **210** components are inserted into mold **300**, the mold **300** may be closed and activated to apply heat and/or pressure as is known in the art. It will be appreciated that the heated adhesive/bonding material will be at least partially expressed from the stack **282** to engage the magnets and the rotor ring, to bond the magnets in the magnet pockets and to bond the rotor ring to the distal ends of the rotor fingers. In this regard, all of the magnets are bonded simultaneously to the rotor core, which significantly reduces the period of time require for manufacture of rotor assembly **210**. By bonding all of the magnets to the rotor core at the same time, drawbacks of the prior art are overcome, particularly where it is found in the prior art that each magnet is inserted separately into a magnet pocket and secured.

[0075] In step **416**, the rotor assembly **210** may be allowed to cure in the mold or may be removed from the mold for curing as is known in the art. It will be appreciated that upon curing, in one or more embodiments, the hardened bonding material may secure the magnets **244** in a spaced apart relationship to the fingers **248** of the rotor core **242**.

[0076] Turning to FIG. **5**, another embodiment of rotor assembly **110** of FIG. **1** is shown and generally designated as rotor assembly **310**. Rotor assembly **310** is generally formed of a rotor core or yoke **342** disposed to carry a plurality of magnets **344**. In one or more embodiments, rotor core **342** is spider shaped and formed of a hub **346** from which fingers **348** radially extend. Fingers **348** are spaced apart from one another around the perimeter of hub **346** so as to form magnet pockets **350** between adjacent fingers **348**. Rotor assembly **310** further includes a rotor ring **352** disposed radially outward from fingers **348**. In one or more embodiments, rotor ring **352** and rotor core **342**



may be separate components, such as is shown in FIG. 5, while in other embodiments, rotor ring 352 and rotor core 342 may be integrally formed.

[0077] In one or more embodiments, rotor core 342 is formed of a non-conducting composite material. Similarly, in one or more embodiments, rotor ring 352 is formed of a non-conducting composite material.

[0078] In the illustrated embodiment of rotor assembly 310, fingers 348 are generally wedge shaped so that magnet pockets 350 are generally rectangular in shape. As such, in this illustrated embodiment, magnets 344 are generally rectangular in shape. This arrangement is desirable because it reduces machining required of the magnets 344 and the magnets can be more readily inserted into the magnet pockets 350. In this regard, this arrangement is more readily produced from the methods described above with respect to FIGS. 2, 3 and 4 than traditional methods because traditional methods of machining would create a greater stress at the proximal end of finger 348 during manufacture, whereas a rotor assembly 310 formed in a mold, such as mold 300 described above, would substantially eliminate such stresses during manufacture.

[0079] Turning to FIGS. 6A-6D with ongoing reference to FIG. 5, magnets 344 may be inserted into magnet pockets 350 in a variety of arrangements. In FIG. 6A, magnet 344 is inserted into the magnet pocket 350 between adjacent fingers 348 of rotor core 342. Rotor ring 352 is positioned radially outward from magnet 344. Magnet 344 includes a radial outer edge 360, a radial inner edge 362 and side edges 364 joining the outer edge 360 and the inner edge 362. In one or more embodiments, one or both of radial outer edge 360 and radial inner edge 362 may be arcuate in shape. In the embodiments of FIGS. 6A-6D, side edges 364 are parallel thereby creating the generally rectangular shape of magnet 344.

[0080] In FIG. 6B, a spacer or wedge 345 is shown disposed in magnet pocket 350 between rotor ring 352 and magnet 344. It has been found that where magnets 344 contact rotor ring 352 directly, undesirable stresses occur at the corners of magnets 344 where the magnets 344 abut the rotor ring 352. By employing spacers 345, the stresses occurring in the rotor assembly 310 at these points are significantly reduced. Spacer 345 may be formed of a variety of materials. In one or more embodiments, spacers 345 may be formed of a metallic or conducting material, while in other embodiments, spacers 345 may be formed of a composite material or a non-conducting material, such as polymers. In yet other embodiments, spacer 345 may be formed of the same or similar magnetic material as magnets 344.

[0081] In FIG. 6C, magnet 344 is positioned in magnet pocket 350 so as to be spaced apart from the fingers 348 defining magnet pocket 350 such that an open space or gap 347 is formed about magnet 344 and each of the sides of rotor core 342 (as defined by the fingers 348 and the hub 346). This allows adhesive/bonding material expressed during the molding process as described above to flow into the open space 347 to better bond to the magnet 344. In one or more embodiments, to further enhance bonding, one or more edges 364 of magnet 344 may include a groove or recess 343a into which the adhesive/bonding material may flow. Alternatively, or in addition thereto, one or more edges 368 of fingers 348 may also include a groove or recess 343b into which the adhesive/bonding material may flow. Any of the grooves 343 as described in FIG. 6C may extend fully or partially such as is generally described above with respect to FIG. 2B. In any event, it will be appreciated that when a rotor assembly 310 having the features described in FIG. 6C is manufactured as described above in manufacturing process 400, the magnets 344 are positioned in mold 300 so that the magnets 344 are spaced apart (see gap 347) from the finger 348 edges and from abutment with the rotor hub 346, thereby permitting a uniform flow of bonding material therebetween. This will reduce the stress on magnets 244 during the high revolution per minute (RPM) operation of rotor assembly 310.

[0082] In FIG. 6D, a connector or fastening mechanism 353, such as a pin or blade, may be inserted into adjacent grooves 343a, 343b of a magnet 344 and a finger 348, respectively, to thereby secure the magnet 344 in the magnet pocket 350 but spaced apart from the fingers 348 of

rotor core **342**.

[0083] FIG. 7 is an enlarged elevation of wedge-shaped spider fingers **348** of the rotor core **342** shown in FIG. 5. To maximize the total surface area of magnets **344** about the perimeter of rotor core **342**, the proximal end **348a** of finger **348** is substantially narrower than the distal end **348b** of finger **348** as illustrated by the respective widths **W1** and **W2**. Forming rotor core **342** by traditional methods of machining would create undesirable stress on the proximal ends **348a** of the fingers **348** during machining. Thus, in one or more embodiments, such a rotor core **342** is formed by stacking a multiplicity of fiber material layers **280** as described above, eliminating the need for machining.

[0084] Turning to FIG. 8, an elevation view of a rotor assembly **310**, similar to FIG. 5 is shown, but with spacers **345** inserted in magnet pockets **350** between the magnets **344** and the rotor ring **352**. As described above with respect to FIG. 6B, spacers **345** absorb the stresses that would otherwise be transferred from rotor ring **352** during operation of rotor assembly **310**. It has been found that such stresses can reduce the operational life of magnets **344** and impact the electrical output of axial flux motors during operation. An aperture **305** may be formed in central hub **346** for receipt of a rotor shaft (see FIG. 1).

[0085] FIGS. 9A-9E illustrate another embodiment of a rotor assembly **500** in accordance with this disclosure. In FIG. 9A, rotor assembly **500** includes a spider shaped rotor core **502** formed along a rotor axis **503** and having a central hub **504** from which fingers **506** radially extend. An aperture **505** may be formed in central hub **504** for receipt of a rotor shaft (see FIG. 1). Fingers **506** are spaced apart from one another around the perimeter of hub **504** so as to form magnet pockets **508** between adjacent fingers **506**. Each magnet pocket **508** is disposed to receive a magnet **510**. A rotor ring **514** is disposed radially outward from fingers **506**. Although not limited to a particular number of fingers **506** and pockets **508**, in the illustrated embodiment, rotor core **502** includes ten fingers **506** and ten magnet pockets **508**. In one or more embodiments, magnets **510** may be positioned in magnet pockets **508** so as to be spaced apart from adjacent fingers **506** and/or central hub **504** so as to form a gap **507** therebetween.

[0086] In one or more embodiments, rotor core **502** may be formed of a plurality of fiber layers bonded together in a fiber layer stack as generally described above, while in other embodiments, rotor core **502** may be formed in other ways.

[0087] FIG. 9B is an exploded view of rotor assembly **500** where it can be seen that each magnet **510** may be supported in a magnet pocket **508** by connectors **513**.

[0088] In the illustrated embodiments, fingers **506** are generally rectangular in shape forming magnet pockets **508** that are generally wedge shaped. As such, in this illustrated embodiment, magnets **510** are generally wedge shaped, where each magnet **510** has a radial outer edge **512a** of a greater length than a radial inner edge **512b** which radial edges **512a**, **512b** are joined by side edges **516**. The radial outer edge **512a** and/or the radial inner edge **512b** may be arcuate. In one or more embodiments, at least one and preferably each magnet side edge **516** includes a magnet groove or slot **518** extending along at least a portion of the length of side edge **516** between radial outer edge **512a** and radial inner edge **512b**. In the illustrated embodiment, each groove **518** fully extends between radial outer edge **512a** and radial inner edge **512b**. Magnet groove **518** is not limited to a particular shape but in some embodiments may be arcuate or rectangular in cross-section.

[0089] To reduce stress on magnets **510**, the outer corners **519** of each magnet **510** may also be rounded.

[0090] Similarly, each finger **506** has a side edge **520** that extends from a proximal finger end **506a** adjacent the hub **504** to a distal finger end **506b** at the periphery of the rotor core **502**. In the illustrated embodiments, each side edge **520** of a finger **506** includes a finger groove or slot **522** extending along at least a portion of the length of side edge **520** between the proximal finger end **506a** and the distal finger end **506b**. Finger groove **522** is not limited to a particular shape but in some embodiments may be arcuate or rectangular in cross-section. Moreover, in one or more embodiments finger groove **522** has the same cross-sectional shape as magnet groove **518** so as to

mirror one another.

[0091] Fingers **506** may be characterized as having a thickness  $T_f$  and magnets **510** may be characterized as having a thickness  $T_m$ . In one or more embodiments, finger thickness  $T_f$  and magnets thickness  $T_m$  are the same. In yet other embodiments, finger thickness  $T_f$  and magnets thickness  $T_m$  may differ to allow formation of cooling channels to be formed in rotor assembly **500**.

[0092] In one or more embodiments, rotor ring **514** is a pre-formed ring that is positioned around rotor core **502**. Rotor ring **514** has a radial inner edge **526** and a radial outer edge **528**, where radial inner edge **526** may abut fingers **506** when rotor ring **514** is positioned around rotor core **502**.

[0093] Connector **513** is disposed to engage each of finger groove **522** and magnet groove **518**. Although connector **513** is not limited to a particular shape, where finger groove **522** and magnet groove **518** are each semi-circular in cross-section, then connector **513** may be a round pin, such as is shown. Likewise, where finger groove **522** and magnet groove **518** are each rectangular in cross-section, then connector **513** may be a square pin or rectangular blade.

[0094] Turning to FIGS. **9C**, **9D** and **9E**, rotor assembly **500** is shown partially assembled, but with rotor ring **514** removed in FIGS. **9D** and **9E** thereby illustrating the positioning of magnets **510** in magnet pockets **508**. Each magnet **510** is positioned in a magnet pocket **508** so that finger grooves **522** align with magnet grooves **518** permitting a connector **513** to be inserted therebetween. In one or more embodiments, magnets **510** are positioned in magnet pockets **508** so as to be spaced apart from adjacent fingers **506** such that a gap **530** is formed between each magnet edge **516** and each finger side edge **520**. It has been found that by holding magnet **510** in spaced apart relationship to fingers **506**, stresses on magnets **510** are reduced, particularly during the high RPMs to which rotor assembly **500** is subjected to during operation. Regardless of whether magnets **510** abut fingers **506** or are spaced apart from fingers **506**, a connector **513** is inserted into the aligned grooves **522**, **518** to secure magnet **510** within magnet pocket **508**. Where a gap **530** is formed, connectors **513** are also utilized to maintain the spaced apart relationship between each magnet **510** the adjacent fingers **506** forming a finger pocket **508** in which the magnet **510** is positioned.

[0095] Although the embodiments of FIGS. **9A-9E** are shown with connectors **513**, in other embodiments, connector **513** can be eliminated. In such case, it will be appreciated that during manufacturing, an adhesive or bonding material may flow into the finger groove **522** and magnet groove **518**, as well as any gap **530**, and upon curing and hardening of the adhesive or bonding material, the magnets **510** are secured in pockets **508** by the hardened material.

[0096] FIGS. **10A** and **10B** illustrate another embodiment of a rotor assembly **600**. In this embodiment, rotor assembly **600** includes a spider shaped rotor core **602** formed along a rotor axis **603** and having a central hub **604** from which fingers **606** radially extend. Although not necessary, in this embodiment, rotor core **602** is shown as having a plurality of fiber layers **605** formed together in a fiber layer stack **609** as generally described above. Fingers **606** are spaced apart from one another around the perimeter of hub **604** so as to form magnet pockets **608** between adjacent fingers **606**. Each magnet pocket **608** is disposed to receive a magnet **610**. A rotor ring **614** is disposed radially outward from fingers **606**. In addition, an outer radial element **611** is positioned as spacers between each magnet **610** and the rotor ring **614**, much in the same way as described above with respect to FIGS. **6B** and **8**.

[0097] Each magnet **610** has a radial outer edge **612a** of a greater length than a radial inner edge **612b** which radial edges **612a**, **612b** are joined by side edges **616**. In one or more embodiments, at least one and preferably each magnet side edge **616** includes a magnet groove or slot **618** extending along at least a portion of the length of side edge **616**. Magnet groove **618** is not limited to a particular shape but in some embodiments may be arcuate or rectangular in cross-section.

[0098] Similarly, each finger **606** has a side edge **620** that extends from a proximal finger end **606a** adjacent the hub **604** to a distal finger end **606b** at the periphery of the rotor core **602**. In the illustrated embodiments, the side edges **620** of fingers are smooth, but in other embodiments, may

include a finger groove or slot as described in FIG. 9.

[0099] In one or more embodiments, rotor ring **614** is a pre-formed ring that is positioned around rotor core **602**. Rotor ring **614** has a radial inner edge **626** and a radial outer edge **628**, where radial inner edge **626** abuts outer radial element **611** when rotor ring **614** is positioned around rotor core **602**.

[0100] A connector **613** is disposed to seat in each magnet groove **618**. In this embodiment, connector **613** may be semicircular in cross-section with an arcuate face that engages groove **618** and a flat face that abuts the smooth edge **620** of finger **606**.

[0101] Outer radial elements **611** are elongated with a first end **615** and a second end **617**, and generally arcuate in shape to engage the radial inner edge **626** of rotor ring **614** as specifically shown in FIG. 10B. One or both ends **615**, **617** may also include a notch **619** to allow a connector **613** to pass therethrough. In one or more embodiments, in addition to securing magnet **610** as described above, connector **613** may also secure outer radial element **611** within a magnet pocket **608** between a magnet **610** and the rotor ring **614**.

[0102] FIGS. 11A and 11B illustrate another embodiment of a rotor assembly **700** in assembled and exploded, respectively. In this embodiment, as shown in FIG. 11A, rotor assembly **700** includes a spider shaped rotor core **702** formed along a rotor axis **703** and having a central hub **704** from which fingers **706** radially extend. Fingers **706** are spaced apart from one another around the perimeter of hub **704** so as to form magnet pockets **708** between adjacent fingers **706**. Each magnet pocket **708** is disposed to receive a magnet **710**. A rotor ring **714** is disposed radially outward from fingers **706**. Each magnet **710** is supported in a magnet pocket **708** by at least one connector assembly **723** positioned between the magnet **710** and the rotor ring **714**.

[0103] In one or more embodiments, rotor core **702** may be formed of a plurality of fiber layers bonded together in a fiber layer stack as generally described above, while in other embodiments, rotor core **702** may be formed in other ways.

[0104] With reference to FIG. 11B and ongoing reference to FIG. 11A, each magnet **710** has a radial outer edge **712a** and a radial inner edge **712b** which radial edges **712a**, **712b** are joined by side edges **716**. In one or more embodiments, at least one and preferably each magnet side edge **716** includes a magnet groove or slot **718** extending along at least a portion of the length of side edge **716**. Magnet groove **718** is not limited to a particular shape but in some embodiments may be arcuate or rectangular in cross-section. In the illustrated embodiment, the radial side edges **716** of a magnet are substantially parallel so that each magnet **710** is generally rectangular in shape. In addition, one or both radial edges **712a**, **712b** may be arcuate as illustrated.

[0105] Similarly, each finger **706** has a side edge **720** that extends from a proximal finger end **706a** adjacent the hub **704** to a distal finger end **706b** at the periphery of the rotor core **702**. Each finger **706** is generally triangular or wedge-shaped, such as discussed above with respect to FIG. 7. As such, the proximal end **706a** of a finger **706** is substantially narrower in width than the distal end **706b** of finger **706** as is also illustrated above in FIG. 7. It will be appreciated that because each finger **706** is wedged shaped, when arranged around rotor core **702**, each set of adjacent wedged shaped fingers **706** that define a magnet pocket **708** have opposing side edges **720** that are substantially parallel such that the magnet pocket **708** formed by the adjacent fingers **706** is substantially rectangular for receipt of a substantially rectangular magnet **710**.

[0106] In the illustrated embodiments, each side edge **720** of a finger **706** includes a finger groove or slot **722** extending along at least a portion of the length of side edge **720** between the proximal finger end **706a** and the distal finger end **706b**. Finger groove **722** is not limited to a particular shape but in some embodiments may be arcuate or rectangular in cross-section. Moreover, in one or more embodiments finger groove **722** has the same cross-sectional shape as magnet groove **718** so as to mirror one another.

[0107] Each connector assembly **723** includes an outer radial element **711** from which extends two connectors **713** so as to be parallel to one another. A connector **713** is disposed to engage the

opposing finger groove **722** and magnet groove **718** of a magnet pocket **708**. Although connector **713** is not limited to a particular shape, where an opposing finger groove **722** and magnet groove **718** are each semi-circular in cross-section such as is shown in FIG. **11A**, then connector **713** may be a round pin, such as is shown. Likewise, where an opposing finger groove **722** and magnet groove **718** are each rectangular in cross-section, then connector **713** may be a square pin or rectangular blade.

[0108] As best seen in FIGS. **11C** and **11D**, in the illustrated embodiment, the two connectors **713a**, **713b** of a connector assembly **723** are disposed to extend from outer radial element **711**. More specifically, each outer radial element **711** includes two spaced apart, substantially parallel connectors **713a**, **713b** extending from the outer radial element **711**, thereby forming a connector assembly **723**. Each outer radial element **711** is elongated, with a first end **715** and a second end **717**, and an inner surface **719** and an outer surface **721**, where one or both surfaces, **719**, **721** in some embodiments may be generally arcuate in shape. In such embodiments, an arcuate outer surface **721** may be disposed to engage the radial inner edge **726** of rotor ring **714** as specifically shown in FIG. **11B**. The first and second connectors **713a**, **713b** of a connector assembly **723** are parallel to one another and extend from the inner surface **719** of an outer radial element **711**. In this embodiment, first and second connectors **713a**, **713b** are spaced inwardly from the first and second ends **715**, **717**, respectively, of the outer radial element **711**.

[0109] Turning FIG. **11E**, rotor assembly **700** is shown partially assembled, but with rotor ring **714** removed thereby illustrating the positioning of magnets **710** in magnet pockets **708**. Each magnet **710** is positioned in a magnet pocket **708** so that finger grooves **722** align with magnet grooves **718**. In one or more embodiments, magnets **710** are positioned in magnet pockets **708** so as to be spaced apart from adjacent fingers **706** such that a gap **730** is formed between each magnet edge **716** and each finger side edge **720**. It has been found that by holding magnet **710** in a spaced apart relationship to finger **706**, stresses on magnets **710** are reduced, particularly during the high RPMs to which rotor assembly **700** is subjected to during operation. Regardless of whether magnets **710** abut fingers **706** or are spaced apart from fingers **706**, a set of connectors **713a**, **713b** of a connector assembly **723** are inserted into the aligned grooves **722**, **718** to secure the magnet **710** within a magnet pocket **708**. Where a gap **730** is formed, connectors **713a**, **713b** are also utilized to maintain the spaced apart relationship between each magnet **710** the adjacent fingers **706** forming a finger pocket **708** in which the magnet **710** is positioned.

[0110] As best seen in FIGS. **11A** and **11C**, in the illustrated embodiment the length of each outer radial element **711** is selected so that it overlays the outer radial edge **712** of a magnet **710** a portion of the distal finger end **706b** of two adjacent fingers **706** that form a magnet pocket **708**.

[0111] FIGS. **12A**, **12B** and **12C** illustrate another embodiment of rotor assembly **700** and connector assemblies **723** shown in FIGS. **11A-11D**, but with connectors **713** being a square pin to engage rectangular shaped finger grooves **722** and magnet grooves **718**.

[0112] FIGS. **12C** and **12D** illustrate another embodiment of rotor assembly **700** and connector assemblies **723** shown in FIGS. **12A** and **12B**, but where each finger **706** includes a notch **725** formed in the side edge **720** at the distal finger end **706b**. In such case, the length of each outer radial element **711** is selected so that connector assembly **725** seats in opposing notches **725** and fits between opposing fingers **706** forming a magnet pocket **708**. In this embodiment, first and second connectors **713a**, **713b** are adjacent the first and second ends **715**, **717**, respectively, of the outer radial element **711**.

[0113] FIGS. **13A-13D** illustrate another embodiment of a rotor assembly **800**. In FIGS. **13A** and **13B**, rotor assembly **800** includes a spider shaped rotor core **802** formed along a rotor axis **803** and having a central hub **804** from which fingers **806** radially extend. In one or more embodiments, rotor core **802** may be formed of a plurality of fiber layers **807** compressed together in a fiber layer stack **809** as generally described above, while in other embodiments, rotor core **802** may be formed in other ways, such as machining. An aperture **805** may be formed in central hub **804** for receipt of

a rotor shaft (see FIG. 1). Fingers **806** are wedge or triangular in shaped and spaced apart from one another around the perimeter of hub **804** so as to form trapezoidal shaped magnet pockets **808** between adjacent fingers **806**. Each magnet pocket **808** is disposed to receive a trapezoidal shaped magnet **810**. A rotor ring **814** is disposed radially outward from fingers **806**.

[0114] FIG. 13C is an exploded view of rotor assembly **800** where it can be seen that each magnet **810** may be supported in a magnet pocket **808** by at least one connector **813**. In other embodiments of rotor assembly **800**, connector **813** may be eliminated, as generally described above.

[0115] Each magnet **810** has a radial outer edge **812a** and a radial inner edge **812b** which radial edges **812a**, **812b** are joined by side edges **816**. The radial inner edge **812b** of a magnet **810** has a greater length than the radial outer edge **812a** such that the magnet **810** is trapezoidal in shape. In one or more embodiments, at least one and preferably each magnet side edge **816** includes a magnet groove or slot **818** extending along at least a portion of the length of side edge **816**. Magnet groove **818** is not limited to a particular shape but in some embodiments may be arcuate or rectangular in cross-section. In the illustrated embodiment, each radial side edges **816** of a magnet angles or tapers inward from the radial inner edge **812b**. In addition, one or both radial edges **812a**, **812b** may be arcuate as illustrated.

[0116] In one or more embodiments, magnets **810** may be positioned in magnet pockets **808** so as to be spaced apart from adjacent fingers **806** and/or central hub **804** so as to form a gap therebetween as generally described above.

[0117] Each finger **806** has a side edge **820** that extends from a proximal finger end **806a** adjacent the hub **804** to a distal finger end **806b** at the periphery of the rotor core **802**. Each finger **806** is generally triangular or wedge-shaped. As such, the proximal end **806a** of a finger **806** is substantially narrower in width than the distal end **806b** of finger **806**. In the illustrated embodiment, the side edges **820** of each finger **806** angle outwardly from proximal end **806a** to define a magnet pocket **808** that is trapezoidal in shape, being wider at the base of the magnet pocket **808** adjacent hub **804** and narrower adjacent the distal end **806b** of fingers **806**. In the illustrated embodiments, the side edges **820** of fingers are smooth, but in other embodiments, may include a finger groove or slot as described in FIG. 9.

[0118] Connector **813** is disposed to seat in each magnet groove **818**. In this embodiment, connector **813** may be semicircular in cross-section with an arcuate face that engages groove **818** and a flat face that abuts the smooth edge **820** of finger **806**.

[0119] It will be appreciated that because magnets **810** are trapezoidal in shape, magnet pockets **808** are likewise trapezoidal in shape so that each magnet pocket **808** is wider at a proximal end **808a** and narrower at a distal end **808b** (as opposed to the rectangular shaped magnet pockets shown in FIGS. 11A-11E).

[0120] In any event, because of their trapezoidal shape, magnets **810** may be secured in magnet pockets **808** by fingers **806** without the need for connectors **813** or even a rotor ring **813** to secure magnets **810** within magnet pockets **808**. Thus, as described above, in some embodiments of rotor assembly **800**, connectors **813**, and even magnet grooves **818** may be eliminated. Likewise, for this reason, rotor ring **814** may be eliminated.

[0121] While unnecessary to secure magnets **810** within magnet pockets **808**, in some embodiments, rotor ring **814** may be provided to strength rotor assembly **800** and in particular to provide support for fingers **806**. In such embodiments, magnets **810** may be sized so that the radial outer edge **812a** of a magnet **810** is spaced apart from the radial inner edge **826** of rotor ring **814** so as to form a radial gap between the magnet **810** and the rotor ring **814**, reducing the likelihood of stress on magnets **810** from rotor ring **814** during operation. In other embodiments, rotor ring **814** may abut the radial outer edge **812a** of magnet **810**.

[0122] FIG. 14A is an exploded view of another embodiment of a rotor assembly **900** where rotor assembly **900** includes a spider shaped rotor core **902** disposed about an axis **903** and having a central hub **904** from which fingers **906** radially extend. In one or more embodiments, rotor core

**902** may be formed of a plurality of fiber layers **907** bonded together in a fiber layer stack **909** as generally described above, while in other embodiments, rotor core **902** may be formed in other ways. Fingers **906** are spaced apart from one another around the perimeter of hub **904** so as to form magnet pockets **908** between adjacent fingers **906**. Each magnet pocket **908** is disposed to receive a magnet **910**. A retaining cover assembly **915** having a first plate **917a** and a second plate **917b** may be utilized to axially enclose and secure rotor core **902** and magnets **910** axially relative to one another.

[0123] Each plate **917** is circular and includes an outer radial edge **919**, an outer surface **921** and an inner surface **923**. Retaining cover assembly **915** may be utilized in one or more of the rotor assembly embodiments described above to provide axial support to rotor core **902** and magnets **910** along axis **903**. In some embodiments of rotor assembly **900**, a rotor ring such as described above may be disposed radially outward of fingers **906** and cooperate with retaining cover assembly **915**. In this regard, in some embodiments, the opposing plates **917a**, **917b** also may axially enclose the rotor ring to provide axial support to the rotor ring. In other embodiments, the outer radial edge **919** of the plates **917a**, **917b** abuts the inner radial edge of the rotor ring. In the embodiment shown in FIG. **14A**, each plate **917** includes an outer radial lip **925** integrally formed on the inner surface **923** of plate **917** adjacent the outer radial edge **919**. In such embodiments, the outer radial lip **925** of each plate **917** may replace the rotor ring described in other embodiments above. In this regard, as best seen in FIG. **14B**, when rotor assembly **900** is assembled, retaining cover assembly **915** fully encloses rotor core **902** and magnets **910** both in the axial and radial directions, and specifically, the radial lip **925** of the two opposing plates **917a**, **917b** abut one another.

[0124] In one or more embodiments, rotor assembly **900** includes internal rotor cooling channels **927** formed to allow air to flow between magnets **910**. In some embodiments, cooling channels **927** may be formed by providing seats **929** on the inner surface **923** of one or both plates **917a**, **917b**.

[0125] In embodiments of rotor assembly **900**, each magnet **910** may be supported in a magnet pocket **908** in a variety of different ways as generally described above, whether with connectors or the shape of fingers **907** or with a rotor ring or even with radial lips **925**, and embodiments of rotor assembly **900** with internal rotor cooling channels **927** is not limited in this regard.

[0126] Each magnet **910** has a radial outer edge **912a** and a radial inner edge **912b** which radial edges **912a**, **912b** are joined by side edges **916**. In one or more embodiments, at least one and preferably each magnet side edge **916** includes a magnet groove or slot **918** extending along at least a portion of the length of side edge **916**.

[0127] Similarly, each finger **906** has a side edge **920** that extends from a proximal finger end **906a** adjacent the hub **904** to a distal finger end **906b** at the periphery of the rotor core **902**. Each side edge **920** of a finger **906** may include a finger groove or slot (not shown in FIG. **14A** but described above in various embodiments) extending along at least a portion of the length of side edge **920** between the proximal finger end **906a** and the distal finger end **906b**.

[0128] A connector (not shown in FIG. **14A** but as described above in various embodiments) may be disposed to engage the opposing finger groove and magnet groove **918**.

[0129] Formed on the inner surface **923** of a plate **917** are one or more raised seats **929** supporting rotor core **902**. In one or more embodiments, a plurality of spaced apart raised seats **929** are formed around axis **903**. Each raised seat **929** is spaced apart from an adjacent seat **929** to partially define a cooling channel **927** therebetween. In the illustrated embodiment, a plurality of wedge-shaped raised seats **929** are formed on the inner surface **923** of plate **917** centrally about axis **903** and extend radially outward towards outer radial edge **919**, thereby defining a plurality of cooling channels **927**. In this embodiment, the raised seats **929** are also spaced apart from outer radial edge **919** of plate **917**. Rotor core **902** is positioned on the raised seats **929** so that the central hub **904** engages the raised seats **929** and fingers **906** align with the cooling channels **927** so that each finger **906** extends along a cooling channel **927**. Thus, in some embodiments, each raised seat **929** is formed to extend radially outward no more than the outer diameter of central hub **904** when central

hub **904** is seated against plate **917**. In any event, when rotor core **902** abuts raised seats **929** as described, fingers **906** are spaced apart from the inner surface **923** of plate **917** such that the cooling channels **927** continue to extend outward along the length of fingers **906**. In one or more embodiments, the seats **929** may be wedge-shaped and positioned around an aperture **933** formed in plate **917**.

[0130] In one or more embodiments, as best seen in FIGS. **14C** and **14D**, magnet thickness  $T_m$  is greater than finger thickness  $T_f$  so that the cooling channel **927** defined along a finger **906** extends between successive magnets **910** when magnets **910** are positioned radially outward of raised seats **929** in magnet pockets **908** formed between adjacent fingers **906**. It will be appreciated that in such an embodiment, a cooling channel **927** extends along a portion of each side edge **916** of a magnet **910**. Where a plate **917** includes a radial lip **925**, each magnet **910** may be positioned and secured radially between a raised seat **929** and the radial lip **925**. In such case, magnet **910** abuts the inner surface **923** of plate **917**. Where both plates **917a**, **917b** include raised seats **929** as described, rotor assembly **900** defines four cooling channels **927** around each magnet **910**. Finally, in one or more embodiments, cooling channels **927** are radially extending. Alternatively, or in addition thereto, cooling channels may also be circumferentially extending channels **931** formed in seat **929**. Such circumferentially extending channels **931** may interconnect radially extending cooling channels. While not shown, in other embodiments, additional seats **929** may be formed on the interior surface **923** of a plate **917** adjacent the distal end **906b** of each finger to provide additional support to a finger **906**.

[0131] Thus, a rotor assembly for an axial flux motor has been described. In one or more embodiments, the rotor assembly generally includes a rotor hub with a plurality of radially extending, spaced apart fingers with adjacent fingers forming a magnet pocket therebetween; a magnet positioned in each magnet pocket of the rotor core; and a rotor ring extending around the periphery of the rotor core, wherein the rotor core comprises a plurality of fiber layers formed together in a fiber layer stack, with each fiber layer having a set of primary fibers extending in a select direction and wherein the sets of primary fibers of adjacent fiber layers extend in different directions. In other embodiments, the rotor assembly may include a rotor core having a rotor hub with a plurality of radially extending, spaced apart fingers with adjacent fingers forming a magnet pocket therebetween; a magnet positioned in each magnet pocket of the rotor core; and a rotor ring extending around the periphery of the rotor core, wherein the rotor core comprises a plurality of material layers formed together in a material layer stack. Likewise, an axial flux motor has been described. In one or more embodiments, the axial flux motor may include at least one stator assembly disposed along an axis; a rotor assembly disposed along the axis and spaced apart from the at least one stator assembly, the rotor assembly comprising a rotor core having a rotor hub with a plurality of radially extending, spaced apart fingers with adjacent fingers forming a magnet pocket therebetween; a magnet positioned in each magnet pocket of the rotor core; and a rotor ring extending around the periphery of the rotor core, wherein the rotor core comprises a plurality of fiber layers formed together in a fiber layer stack, with each fiber layer having a set of primary fibers extending in a select direction and wherein the sets of primary fibers of adjacent fiber layers extend in different directions. In other embodiments, the rotor assembly generally includes a rotor core having a rotor hub with a plurality of radially extending, spaced apart fingers forming a magnet pocket between adjacent fingers; a magnet positioned in each magnet pocket of the rotor core; wherein the rotor core comprises a plurality of fiber layers formed together in a fiber layer stack, with each fiber layer comprises a material having a set of primary fibers extending in a select direction and wherein the sets of primary fibers of adjacent fiber layers extend in different directions. In yet other embodiments, the rotor assembly may include a rotor core having a rotor hub with a plurality of radially extending, spaced apart fingers forming a magnet pocket between adjacent fingers; a magnet positioned in each magnet pocket of the rotor core; wherein the rotor core comprises a plurality of fiber layers formed together in a fiber layer stack. In yet other



embodiments, the rotor assembly may include a rotor core having a rotor hub with a plurality of radially extending, spaced apart fingers forming a magnet pocket between adjacent fingers; a magnet positioned in each magnet pocket of the rotor core; a rotor ring extending around the periphery of the rotor core; and a spacer positioned radially outward of each magnet between the magnet and the rotor ring. In yet other embodiments, the rotor assembly may include a rotor core having a rotor hub with a plurality of radially extending, spaced apart fingers forming a magnet pocket between adjacent fingers; a magnet positioned in each magnet pocket of the rotor core; and a rotor ring extending around the periphery of the rotor core; wherein each magnet has a radial outer edge and a radial inner edge which radial edges are joined by side edges; and wherein each side edge has a groove formed therein along at least a portion of the length of the side edge. In yet other embodiments, the rotor assembly may include a rotor core having a rotor hub with a plurality of radially extending, spaced apart fingers forming a magnet pocket between adjacent fingers; a magnet positioned in each magnet pocket of the rotor core, wherein each magnet has a radial outer edge and a radial inner edge which radial edges are joined by side edges; and wherein each side edge has a groove formed therein along at least a portion of the length of the side edge; a rotor ring extending around the periphery of the rotor core; and a hardened bonding material between each finger and an adjacent magnet side edge, the hardened bonding material disposed in the groove of the adjacent magnet. In yet other embodiments, the rotor assembly may include a rotor core having a rotor hub with a plurality of radially extending, spaced apart fingers forming a magnet pocket between adjacent fingers, wherein each finger has two finger side edges extending from a proximal finger end adjacent the rotor hub to a distal finger end at a periphery of the rotor core, and wherein each finger side edge has a finger groove formed therein along at least a portion of the length of the finger side edge; a magnet positioned in each magnet pocket of the rotor core, wherein each magnet has a radial outer edge and a radial inner edge which radial edges are joined by magnet side edges; and wherein each side edge has a magnet groove formed therein along at least a portion of the length of the magnet side edge, wherein at least one finger groove of the fingers aligns with at least one magnet groove of the magnets; a rotor ring extending around the periphery of the rotor core; and a connector inserted in the aligned finger groove and magnet groove. In yet other embodiments, the rotor assembly may include a rotor core having a rotor hub with a plurality of radially extending, spaced apart fingers forming a magnet pocket between adjacent fingers, wherein each finger has two finger side edges extending from a proximal finger end adjacent the rotor hub to a distal finger end at a periphery of the rotor core; a magnet positioned in each magnet pocket of the rotor core, wherein each magnet has a radial outer edge and a radial inner edge which radial edges are joined by magnet side edges; and a rotor ring extending around the periphery of the rotor core; wherein the magnet side edge is spaced apart from the finger side edge on each side of each magnet. In yet other embodiments, the rotor assembly may include a rotor core having a rotor hub with a plurality of radially extending, spaced apart, wedge shaped fingers forming a magnet pocket between adjacent fingers, wherein each finger has two finger side edges extending from a proximal finger end adjacent the rotor hub to a distal finger end at a periphery of the rotor core, wherein the finger side edges of adjacent fingers forming a magnet pocket are parallel with one another, and wherein each finger side edge has a finger groove formed therein along at least a portion of the length of the finger side edge; a magnet positioned in each magnet pocket of the rotor core, wherein each magnet has a radial outer edge and a radial inner edge which radial edges are joined by magnet side edges that are parallel to one another, and wherein each magnet side edge has a magnet groove formed therein along at least a portion of the length of the magnet side edge, wherein at least one finger groove of the fingers aligns with at least one magnet groove of the magnets; and a connector assembly adjacent the radial outer edge of each magnet; wherein each connector assembly comprises an outer radial element from which extends a first connector spaced apart from a second connector, wherein the first and second connectors are parallel to one another. In yet other embodiments, the rotor assembly may include a rotor core having a rotor hub with a plurality of

radially extending, spaced apart, wedge shaped fingers forming a magnet pocket between adjacent fingers, wherein each finger has two finger side edges extending from a proximal finger end adjacent the rotor hub to a distal finger end at a periphery of the rotor core; and a magnet positioned in each magnet pocket of the rotor core, wherein each magnet has a radial outer edge of a first width and a radial inner edge of a second width, which radial edges are joined by magnet side edges, where the first width is less than the second width so that the magnet is trapezoidal in shape. In yet other embodiments, the rotor assembly may include a rotor core having a rotor hub with a plurality of radially extending, spaced apart fingers forming a magnet pocket between adjacent fingers, wherein each finger has two finger side edges extending from a proximal finger end adjacent the rotor hub to a distal finger end at a periphery of the rotor core; a magnet positioned in each magnet pocket of the rotor core, wherein each magnet has a radial outer edge and a radial inner edge, which radial edges are joined by magnet side edges; and a cooling channel formed along at least one finger between adjacent magnets. In yet other embodiments, the axial flux motor may include at least one stator assembly disposed along an primary axis; a rotor assembly disposed along the primary axis and spaced apart from the at least one stator assembly; and a cooling mechanism on which the at least one stator assembly is supported, wherein the cooling mechanism comprises a plate having an exterior surface and an interior surface, wherein the stator assembly is attached to the interior surface, and wherein a plurality of fluidically connected cooling channels are formed in the exterior surface. In yet other embodiments, the axial flux motor may include at least one stator assembly disposed along an primary axis; a rotor assembly disposed along the primary axis and spaced apart from the at least one stator assembly; and a cooling mechanism on which the at least one stator assembly is supported, wherein the cooling mechanism comprises a plate having an exterior surface and an interior surface, wherein the stator assembly is attached to the interior surface by fasteners that extend from the stator assembly and engage the interior surface of the plate but are spaced apart from the exterior surface of the plate.

[0132] For any of the foregoing embodiments, the rotor assembly may include any one of the following elements, alone or in combination with each other: [0133] The rotor core has a rotor core perimeter and comprises a hub with a plurality of fingers radially extending from the hub, each finger having a proximal end adjacent the hub and a distal end adjacent the rotor core perimeter, the plurality of fingers spaced circumferentially around the hub so as to be spaced apart from one another so as to form a magnetic pocket between adjacent fingers. [0134] A plurality of the material layers each have a set of primary fibers extending in a select direction. [0135] The cooling mechanism comprises a plate having an exterior surface and an interior surface, wherein a plurality of fluidically connected cooling channels are formed in the exterior surface. [0136] The set of primary fibers comprise glass fibers. [0137] The set of primary fibers comprise carbon fibers. [0138] The rotor core is spider shaped. [0139] The magnet side edge is spaced apart from the finger side edge on each side of the magnet. [0140] The magnet radial inner edge is spaced apart from the rotor hub. [0141] The fasteners attaching the stator assembly extend only partially into the plate. [0142] The plate wherein the stator assembly is attached to the interior surface by fasteners that extend from the stator assembly and engage the interior surface of the plate but are spaced apart from the exterior surface of the plate. [0143] Each connector assembly engages a magnet so that the parallel first and second connectors of the connector assembly extend into aligned magnet and finger grooves and wherein the radial outer element of the connector assembly abuts the radial outer edge of the magnet. [0144] Each connector assembly overlays only a portion of adjacent fingers forming a magnet pocket. [0145] A cooling channel formed along each finger between adjacent magnets. [0146] The cooling channel is radially extending. [0147] A retaining cover assembly having a first plate axially spaced apart from a second plate, wherein each plate is circular and includes an outer radial edge, an outer surface and an inner surface, and wherein at least one plate includes spaced apart seats formed on the inner surface of one of the plates, the

spaced apart seats at least partially forming a cooling channel. [0148] The rotor core is formed of a hub from which fingers radially extend, wherein said fingers are spaced apart from one another around the perimeter of the hub so as to form magnet pockets between adjacent fingers. [0149] The rotor assembly further comprises a rotor ring disposed radially outward from the fingers. [0150] At least ten fingers and at least ten magnet pockets. [0151] A plurality of fingers are each rectangular in shape and a plurality of magnet pocket are each wedge shaped. [0152] A plurality of fingers are each wedge shaped and a plurality of magnet pocket are rectangular in shape. [0153] Each magnet has a radially outer edge of a first length and a radially inner edge of a second length that is less than the first length. [0154] Each magnet has a radially outer edge of a first length and a radially inner edge of a second length that is greater than the first length. [0155] Each magnet has a radially outer edge of a first length and a radially inner edge of a second length that is the same as the first length. [0156] Each magnet has side edges that extend from the radially outer edge to the radially inner edge of the magnet. [0157] A groove extending along at least a portion of the length of a side edge of a magnet between said magnets radially outer edge and radially inner edge. [0158] The groove fully extends between the radially outer edge and the radially inner edge of said magnet. [0159] The groove extends from the radially outer edge and is spaced apart from radially inner edge of said magnet. Said groove is spaced apart from the radially outer edge and extends from radially inner edge of said magnet. [0160] The groove is spaced apart from both the radially outer edge and the radially inner edge of said magnet. [0161] A groove extending along at least a portion of the length of a side edge of a finger between the proximal end and the distal end of said finger. [0162] The groove fully extends between the proximal end and the distal end of said finger. [0163] The groove extends from the distal end and is spaced apart from proximal end of said of said finger. [0164] The groove is spaced apart from the distal end and extends from the proximal end of said finger. [0165] The groove is spaced apart from both the distal end and the proximal end of said finger. [0166] A groove along each side edge of a magnet. [0167] A groove along each side edge of a finger. [0168] The groove is rectangular or square in shape and said connector is a blade. [0169] The groove is semicircular in shape and said connector is a round pin. [0170] A spacer radially positioned between an outer radial edge of a magnet and an outer ring. [0171] Each magnet has a separate spacer radially positioned between an outer radial edge of the magnet and an outer ring. [0172] A plurality of spacers positioned between the outer radial edges of the magnets and an outer ring. [0173] The spacer comprises a spacer body. [0174] The spacer body is arcuate in shape. [0175] The spacer further comprises a connector extending radially inward from said spacer body. [0176] Spacer further comprises spaced apart connectors extending radially inward from said spacer body. [0177] The spaced apart connectors are parallel. [0178] The grooves of adjacent finger edges are parallel. [0179] The grooves of radially extending edges of a magnet are parallel. [0180] An integrally formed rotor ring and rotor core, wherein the integrally formed rotor ring and rotor core comprises a plurality of radially extending fingers spaced apart from one another around the perimeter of a hub and extending to a ring disposed around the core, wherein adjacent fingers form a magnet pocket therebetween. [0181] Each material layer in the material layer stack is shaped to comprises a plurality of radially extending fingers spaced apart from one another around the perimeter of a hub and extending to a ring disposed around the core, wherein adjacent fingers form a magnet pocket therebetween. [0182] A rotor ring disposed radially outward from fingers. [0183] A rotor ring disposed radially outward from the spacers. [0184] The rotor ring is integrally formed with the fingers. [0185] The rotor ring is a continuous ring. [0186] The rotor ring has a radially inner edge and a radially outer edge, where the radially inner edge is adjacent the distal end of the fingers. [0187] The rotor ring is formed of a first material at the radially inner edge and a second material at the radially outer edge. [0188] The first material is glass fiber and the second material is carbon fiber. [0189] Each finger has a proximal width adjacent the proximal end of said finger and a distal width adjacent the distal end of each finger, and the distal width is greater than the proximal width. [0190] Each magnet pocket narrows in a radially outward direction. [0191] The inner radial

edge of each magnet is longer than the outer radial edge of each magnet. [0192] The inner radial edge of each magnet is adjacent the proximal end of a finger and the outer radial edge of each magnet is adjacent the distal end of a finger. [0193] Each magnet pocket is reverse trapezoidal in shape and wherein each magnet is trapezoidal in shape and disposed to seat in a reverse trapezoidal shaped pocket. [0194] Each fiber layer having a set of primary fibers extending in a select direction and wherein the sets of primary fibers of adjacent fiber layers extend in different directions. [0195] A rotor core having a rotor hub with a plurality of radially extending, spaced apart fingers forming a magnet pocket between adjacent fingers; and a magnet positioned in each magnet pocket of the rotor core; [0196] The rotor core comprises a plurality of fiber layers formed together in a fiber layer stack, with each fiber layer comprises a material having a set of primary fibers extending in a select direction and wherein the sets of primary fibers of adjacent fiber layers extend in different directions. [0197] A rotor core having a rotor hub with a plurality of radially extending, spaced apart fingers forming a magnet pocket between adjacent fingers, wherein each finger has two finger side edges extending from a proximal finger end adjacent the rotor hub to a distal finger end at a periphery of the rotor core; [0198] A magnet positioned in each magnet pocket of the rotor core, wherein each magnet has a radial outer edge and a radial inner edge which radial edges are joined by magnet side edges; [0199] A plurality of layers of material arranged into a layer stack, with each layer having a set of primary fibers extending in a select direction that is a different direction from the set of primary fiber layers of an adjacent layer in the layer stack. [0200] Each magnet side edge is spaced apart from each finger side edge to form a gap between each magnet side edge and an finger side edge. [0201] A connector, wherein at least one of the side edges of a magnet or a finger includes a groove formed therein along at least a portion of the length of the side edge; and wherein the connector is inserted into the groove to secure a magnet within a magnet pocket. [0202] A rotor ring extending around a periphery of the rotor core. [0203] A spacer positioned radially outward of each magnet between the magnet and the rotor ring. [0204] A radially extending cooling channel formed along at least one finger between adjacent magnets. [0205] The radial outer edge of each magnet has a first width, and the radial inner edge of each magnet has a second width, where the first width is less than the second width so that each magnet is trapezoidal in shape; and wherein each radially extending, spaced apart fingers forming a magnet pocket is wedge shaped so as to form a corresponding trapezoidal shaped magnet pocket for receipt of a trapezoidal shaped magnet. [0206] The finger side edges of adjacent fingers forming a magnet pocket are parallel with one another; and the magnet side edges of each magnet are parallel to one another. [0207] Each finger side edge has a finger groove formed therein along at least a portion of the length of the finger side edge; and wherein each side edge has a magnet groove formed therein along at least a portion of the length of the magnet side edge, wherein at least one finger groove of the fingers aligns with at least one magnet groove of the magnets. [0208] A cooling channel formed along each finger between adjacent magnets. [0209] Each magnet has a magnet thickness and each finger has a finger thickness, wherein the magnet thickness is greater than the finger thickness. [0210] A plurality of spaced apart seats, wherein the seats are wedge-shaped and positioned around an aperture formed in the plate on which the seats are formed so that the cooling channels extend radially out from a center of the plate. [0211] The rotor core abuts the plate so that the rotor hub engages the seats and each magnet abuts the inner surface of the plate. [0212] Each plate further comprises an outer radial lip on the inner surface.

[0213] Thus, a method for manufacturing a rotor for an axial flux motor has been described. The method generally includes arranging a plurality of fiber layers into a fiber layer stack, with each fiber layer having a set of primary fibers extending in a select direction that is a different direction from the set of primary fiber layers of an adjacent fiber layer in the fiber layer stack; inserting a pre-formed rotor ring into a mold; inserting a plurality of magnets into the mold within the rotor ring; inserting the fiber layer stack into the mold; activating the mold to express bonding material from the fiber layer stack and to bond the rotor ring and magnets to the fiber layer stack. In another

embodiment, the method may generally include arranging a plurality of layers of material into a layer stack, with each layer having a set of primary fibers extending in a select direction that is a different direction from the set of primary fiber layers of an adjacent layer in the layer stack; inserting a pre-formed rotor ring into a mold; inserting a plurality of magnets into the mold within the rotor ring; inserting the fiber layer stack into the mold; and activating the mold to express bonding material from the layer stack to bond the rotor ring and magnets to the layer stack. It yet another embodiment, the method may include cutting a plurality of layers from a material having a primary set of fibers aligned in a first direction; and arranging the layers into a layer stack, with each layer having its primary set of fibers extending in a select direction that is a different direction from an adjacent layer in the layer stack.

[0214] For any of the foregoing embodiments, the method may include any one of the following, alone or in combination with each other: [0215] Applying heat and pressure to the layer stack to express bonding material from the stack to bond the layers to one another. [0216] Inserting a pre-formed rotor ring into a mold; inserting a plurality of magnets into the mold within the rotor ring; inserting the layer stack into the mold; and activating the mold to express bonding material from the layer stack to bond the rotor ring and magnets to the layer stack. [0217] Preheating the rotor ring before inserting it into the mold. [0218] Preheating the magnets before inserting them into the mold.

[0219] Although various embodiments have been shown and described, the disclosure is not limited to such embodiments and will be understood to include all modifications and variations as would be apparent to one skilled in the art. Therefore, it should be understood that the disclosure is not intended to be limited to the particular forms disclosed; rather, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the disclosure as defined by the appended claims.

## Claims

1. A rotor assembly for an axial flux motor comprising: a rotor core formed about a rotor axis and having a rotor hub with a plurality of spaced apart fingers radially extending from the rotor hub and forming a magnet pocket between adjacent fingers, wherein each finger has two finger side edges radially extending from a proximal finger end adjacent the rotor hub to a distal finger end at a periphery of the rotor core; a magnet positioned in each magnet pocket of the rotor core, wherein each magnet has a radial outer edge and a radial inner edge which radial edges are joined by radially extending magnet side edges; and a radially extending cooling channel formed along at least one finger between adjacent magnets.
2. The rotor assembly of claim 1, where each magnet has a magnet thickness and each finger has a finger thickness, wherein the magnet thickness is greater than the finger thickness so as to form the radially extending cooling channel along at least one finger.
3. The rotor assembly of claim 2, further comprising a retaining cover assembly having a first plate axially spaced apart from a second plate, wherein each plate is circular and includes an outer surface and an inner surface, and wherein the magnets bear against at least one plate to enclose the cooling channel along a finger.
4. The rotor assembly of claim 3, wherein the at least one plate includes a plurality of spaced apart seats formed on the inner surface of the at least one plate, wherein each seat has a magnet bearing against the seat.
5. The rotor assembly of claim 1, further comprising a retaining cover assembly having a first plate axially spaced apart from a second plate, wherein each plate is circular and includes an outer surface and an inner surface with at least one seat formed on the inner surface of the at least one plate, and wherein the rotor hub bears against the at least one seat to form the cooling channel along a finger.

**6.** A rotor assembly for an axial flux motor comprising: a rotor core having a rotor hub disposed about a rotor axis, and a plurality of spaced apart fingers radially extending from the hub to form a magnet pocket between adjacent fingers, wherein each finger has two finger side edges extending from a proximal finger end adjacent the rotor hub to a distal finger end at a periphery of the rotor core; and a magnet positioned in each magnet pocket of the rotor core, wherein each magnet has a radial outer edge and a radial inner edge, which radial edges are joined by magnet side edges; and a retaining cover assembly having a first plate axially spaced apart from a second plate, wherein each plate is circular and includes an outer radial edge, an outer surface and an inner surface, and wherein at least one plate includes at least one seat formed on the inner surface of the at least one plate, the at least one seat at least partially forming a radially extending cooling channel adjacent the at least one plate.

**7.** The rotor assembly of claim 6, further comprising a cooling channel formed along each finger between adjacent magnets.

**8.** The rotor assembly of claim 6, where each magnet has a magnet thickness and each finger has a finger thickness, wherein the magnet thickness is greater than the finger thickness.

**9.** The rotor assembly of claim 6, wherein the at least one seat comprises a plurality of wedge-shaped seats positioned around an aperture formed in the at least one plate so that the cooling channels extend radially out from a center of the plate.

**10.** The rotor assembly of claim 6, wherein the rotor hub bears against the at least one seat to form the cooling channel along a finger.

**11.** An axial flux motor comprising: a first end plate with an interior surface and an exterior surface; a first stator assembly disposed along a motor axis and mounted on the interior surface of the first end plate; a second end plate with an interior surface and an exterior surface; a second stator assembly disposed along the motor axis and mounted on the interior surface of the second end plate; a rotor assembly disposed along the motor axis and positioned between the first stator assembly and the second stator assembly, the rotor assembly comprising a rotor core formed about a rotor axis and having a rotor hub with a plurality of spaced apart fingers radially extending from the rotor hub and forming a magnet pocket between adjacent fingers, wherein each finger has two finger side edges radially extending from a proximal finger end adjacent the rotor hub to a distal finger end at a periphery of the rotor core; and a magnet positioned in each magnet pocket of the rotor core, wherein each magnet has a radial outer edge and a radial inner edge which radial edges are joined by radially extending magnet side edges; a motor housing extending between the first end plate and the second end plate to enclose the first stator assembly, the second stator assembly and the rotor assembly, wherein at least one end plate comprises fastener bores formed on the interior surface of the end plate and extending only partially through the end plate from the interior surface; and where the at least one end plate further comprises at least one cooling mechanism formed in the exterior surface of the at least one end plate so as not to intersect with the fastener bores.

**12.** The axial flux motor of claim 11, wherein the at least one cooling mechanism is a cooling channel formed in the exterior surface of the at least one end plate so as not to intersect with the fastener bores.

**13.** The axial flux motor of claim 12, wherein the at least one cooling channel is a plurality of fluidically connected cooling channels.

**14.** The axial flux motor of claim 13, wherein the plurality of cooling channels forms a star shape.

**15.** The axial flux motor of claim 14, wherein the star shaped cooling channels are spaced apart from the fastener bores of the end plate.

**16.** The axial flux motor of claim 12, wherein the at least one cooling channel forms a star shape and is spaced apart from the fastener bores of the end plate.

**17.** The axial flux motor of claim 12, wherein the fastener bores are threaded.

**18.** The axial flux motor of claim 11, further comprising fasteners engaging the fastener bores to

secure the first stator assembly to the interior surface of the first end plate.

**19.** The axial flux motor of claim 11, wherein the at least one cooling mechanism is a cooling chamber formed in the exterior surface of the at least one end plate so as not to intersect with the fastener bores.

**20.** The axial flux motor of claim 11, wherein the at least one cooling mechanism is a plurality of cooling ribs formed in the exterior surface of the at least one end plate so as not to intersect with the fastener bores.

**21.** The axial flux motor of claim 11, further comprising an outer plate disposed over the cooling mechanism formed in the exterior surface of the at least one end plate so as to enclose the cooling mechanism.

**22.** The axial flux motor of claim 21, further comprising one or more ports in fluid communication with the cooling mechanism.

**23.** An axial flux motor comprising: a first end plate having an interior surface and an exterior surface; a first stator assembly disposed along a motor axis and mounted on the interior surface of the first end plate; a second end plate having an interior surface and an exterior surface; a second stator assembly disposed along the motor axis and mounted on the interior surface of the second end plate; a rotor assembly rotatably disposed along the motor axis between the first stator assembly and the second stator assembly; a motor housing extending between the first end plate and the second end plate to enclose the first stator assembly, the second stator assembly and the rotor assembly, wherein at least one end plate comprises fastener bores formed on the interior surface of the end plate and extending only partially through the end plate from the interior surface; and where the at least one end plate further comprises at least one cooling channel formed in the exterior surface of the at least one end plate so as to be spaced apart from the interior surface of the at least one end plate.

**24.** The axial flux motor of claim 23, wherein the plurality of cooling channels forms a star shape.

**25.** The axial flux motor of claim 23, further comprising fastener bores formed on the interior surface of the end plate and extending only partially through the end plate from the interior surface so as not to intersect with the at least one cooling channel.

**26.** The axial flux motor of claim 25, wherein the at least one cooling channel forms a star shape and is spaced apart from the fastener bores of the end plate.

**27.** The axial flux motor of claim 26, further comprising outer plate disposed over the cooling mechanism formed in the exterior surface of the at least one end plate so as to enclose the cooling mechanism.

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